(iii) $2x^2 - 2x + 9 = 0$

Here, $a = 2$, $b = -2$, $c = 9$

Now, $\Delta = b^2 - 4ac = \left(-2\right)^2 - 4\left(2\right)\left(9\right) = -68$

Here, $\Delta = -68 < 0$. So, the equation will have no real roots.

Example 3.41 (i) Find the values of 'k', for which the quadratic equation $kx^2 - (8k + 4)x + 81 = 0$ has real and equal roots?

(ii) Find the values of 'k' such that quadratic equation $(k + 9)x^2 + (k + 1)x + 1 = 0$ has no real roots?

Solution (i) $kx^2 - (8k + 4)x + 81 = 0$

Since the equation has real and equal roots, $\Delta = 0$.

That is, $b^2 - 4ac = 0$

Here, $a = k$, $b = -(8k + 4)$, $c = 81$

That is, $\left[-(8k + 4)\right]^2 - 4(k)(81) = 0$

$$64k^2 + 64k + 16 - 324k = 0$$

$$64k^2 - 260k + 16 = 0$$

Dividing by 4 we get $16k^2 - 65k + 4 = 0$

$(16k - 1)(k - 4) = 0$ then, $k = \frac{1}{16}$ or $k = 4$

(ii) $(k + 9)x^2 + (k + 1)x + 1 = 0$

Since the equation has no real roots, $\Delta < 0$

That is, $b^2 - 4ac < 0$

Here, $a = k + 9$, $b = k + 1$, $c = 1$

That is, $(k + 1)^2 - 4(k + 9)(1) < 0$

$$k^2 + 2k + 1 - 4k - 36 < 0$$

$$k^2 - 2k - 35 < 0$$

$$(k + 5)(k - 7) < 0$$

Therefore, $-5 < k < 7$. {If $\alpha < \beta$ and if $(x - \alpha)(x - \beta) < 0$ then, $\alpha < x < \beta$}.

Example 3.42 Prove that the equation $x^2(p^2 + q^2) + 2x(pr + qs) + r^2 + s^2 = 0$ has no real roots. If $ps = qr$, then show that the roots are real and equal.

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